

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Principles of pollution control (CPA, CGM, new technologies) - Set 1

Mark scheme

AQA spec 3.4.3.1, 3.4.3.3; Pollution; textbook pp. 243-245, 303-304

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Predicting the route / movement a pollutant will take through the environment (based on its properties and features of the environment) • To assess where it will become concentrated / which locations and organisms are most at risk, so monitoring can focus on those sites	2
01.2	<i>Any 3 from:</i> • Sea current velocity and direction • Properties of the radionuclides: half-life, solubility, tendency to adsorb onto sediments / bioaccumulate • Food chain pathways, eg uptake by seaweed, shellfish, fish eaten by people • Geology / hydrology of the coast; sediment deposition areas (mud flats) • Human activities in the area: fishing, seaweed collection, beach use	3
01.3	<i>Any 3 from:</i> • Identify the group of people whose lifestyle gives them the highest exposure (eg people who eat a lot of locally caught shellfish / fish, live near the mud flats, collect seaweed) • Monitor their exposure / the concentration of radioactive material in the foods they eat or their body • If their exposure / risk is acceptably low, everyone else is assumed to be at even lower risk; if not, emissions are controlled • CGM assesses exposure before health effects occur (not used to detect damage)	3

Question	Marking guidance	Marks
01.4	<i>Any 2 from:</i> • Seawater • Sediment / mud • Seaweed • Shellfish / molluscs / fish • Milk, vegetables or meat from nearby farms • Atmospheric dust • Soil	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243, 303-304; AQA spec 3.4.3.1.	
02.1	• Location: discharge into the sea / large water bodies where currents dilute and disperse the pollutant • Location: chimneys / emissions sited downwind of urban areas so pollutants are not carried over populations; avoiding valleys and enclosed water bodies where pollutants concentrate • Location: not discharging wastes onto permeable rock above an aquifer / not siting landfill on aquifers • Timing: restrict polluting activities (industrial combustion, vehicle use) during temperature inversions when dispersal is poor • Timing: discharge liquid effluent when river flow is high / on an ebb tide so it is diluted and carried away • Timing: pesticide spraying at night / on still days; noisy operations restricted to daytime; night-flight restrictions	6
02.2	• The organisation / person that causes the pollution is responsible for (paying for) the damage caused • If the cost of preventing the pollution is less than the cost of the damage / fines, there is a financial incentive to prevent it; works best when penalties are high and escape is unlikely	2
02.3	• It is assumed that the waste will cause pollution / harm until research shows it is unlikely to • So release is not permitted until it has been tested; lack of knowledge of a problem is not an excuse / does not reduce responsibility	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
03.1	• Cost increases as the percentage prevented increases • The relationship is not linear: cost rises slowly at first then very steeply / exponentially above about 80-90%	2
03.2	• $16.0 - 5.0 = 11.0$ (£ million)	1
03.3	• ALARA: emissions should be As Low As Reasonably Achievable • It is rarely practical / affordable to reduce emissions to zero; low-level emissions may have acceptable impacts so an acceptable emission level is set • BATNEEC: Best Available Technology Not Entailing Excessive Cost - choosing the most effective control equipment whose cost is not excessive • The factory would install equipment that gives a high level of control (eg 80-90%) rather than paying £16 million for 99%	4

Question	Marking guidance	Marks
03.4	<i>Any 3 from:</i> • Industry / city may have expanded so there are more pollutant sources and total emissions have risen • Better understanding of the health / environmental effects so stricter standards are set • Improved technology becomes available (BATNEEC changes) / cost of better control falls • Pollutant may have accumulated (persistent) so background levels are higher	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244-245; AQA spec 3.4.3.1.	
04.1	• Production prevention: desulfurisation of fuels before combustion; low-temperature (fluidised bed) combustion to prevent NO_x; smokeless fuels • Prevention of release: electrostatic precipitators / bag filters / cyclones for smoke; catalytic converters; flue gas desulfurisation; bund walls; landfill liners • Post-release remediation: oil spill clean-up (booms, skimmers, dispersants, bioremediation); phytoremediation of metal-contaminated land; liming of acidified lakes • Alternative processes: electric vehicles instead of petrol / diesel; renewable energy instead of fossil fuels; pyrethroids instead of organochlorines; fuel cells instead of internal combustion engines	4
04.2	• Remediation is often incomplete / cannot recover all the pollutant (eg oil spills, dispersed pollutants), so some damage still occurs • Prevention avoids the damage entirely and is usually cheaper than clean-up / avoids polluter-pays costs; moving from post-production treatment to non-release is a circular economy principle	2
04.3	• Satellites detect and track the extent / movement of oil slicks over large areas, quickly and repeatedly, so clean-up can be directed and polluters identified • GPS tracking monitors ship positions so navigation errors / collisions / groundings are reduced and offshore shipping routes are enforced • Polymers with a high surface area adsorb metal ions from contaminated water / effluent so they can be removed and recovered • Plants that absorb heavy metals (hyperaccumulators) are grown on contaminated land, harvested and disposed of / metals recovered, cleaning the soil	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
05.1	• C	1
05.2	• Monitoring can be focused on the sites predicted to be most at risk / where the pollutant will concentrate • No need to sample sites that were never likely to be polluted; sampling also checks the accuracy of the predictions	2
05.3	• Behaviour of new chemicals can be predicted from similar existing chemicals: it is likely to persist, bioaccumulate and biomagnify • So acceptable emission limits, control methods and monitoring programmes can be set before release (precautionary principle)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
06.1	• A waste is assumed to cause pollution until research shows it is unlikely to do so	1
06.2	• If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution • If polluters can escape their responsibilities / penalties are low, there is little incentive to comply	2
06.3	<i>Any 2 from:</i> • Marine discharges where currents dilute and disperse emissions • Emissions downwind of urban areas • Not discharging waste onto permeable rock above an aquifer	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.	
07.1	• Natural processes may concentrate the material (eg adsorption onto sediments, bioaccumulation and food-chain concentration) • It may be carried to locations where it is hazardous, such as fishing grounds or beaches used by people	2
07.2	<i>Any 4 from:</i> • Physical state of the effluent (solution / solid) • Ocean currents / river flow • Effect of pH and oxygen on solubility • Bioaccumulation and food-chain concentration • Food sources and consumption by local people (eg shellfish, seaweed) • Half-lives of the isotopes • Meteorological conditions for any gaseous release • Nature of the environment - geology, vegetation	4
07.3	• Monitoring focuses on the sites most at risk of contamination • No need to sample sites that were never likely to be polluted; sampling checks the accuracy of the predictions	2
07.4	• It assesses the risk to the people most exposed because of their lifestyle before any health effects occur • If their risk is acceptably low everyone else has a lower risk; if unacceptable, emissions are reduced to prevent future exposure	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	